Flying to the Past

Contents

Glory Days in Lisbon 2
The Blown Away Kite 6
Icarus 8
Metropolis 12
"Air Farce" 16
Inherited Wings 20
The Fledglings 24
Roaming through 1969 28

Glory Days in Lisbon

CH19601-01

Glory Days in Lisbon

And once you have tasted flight,

You will walk the earth with your eyes turned skyward,

For there you have been,

And there you would return.

Coast, Lisbon, 1835

The air smells of lemon and bay leaves.

The wind roars in from the Atlantic, but no sails wait to catch it. A military blockade sits anchored at the mouth of the harbor. No ship inside is to depart without permission.

It is summer, in the year 1835, and the people of Lisbon, still recovering from years of brutal civil war, are anxious once more. The Queen has banned all passage from the city to the New World.

Above the forbidden sea, only the skies offer a possibility of escape.

Lucy: Are you ready, Jimena?

Jimena: Ready!

Lucy: Three, two, one—

After the mechanical woman finishes her countdown, her able pilot takes a deep breath and opens her arms.

Artificial wings of metal, wood, and leather deploy out from her back. Flapping them like a bird, she runs toward the water and takes a flying leap.

Jimena: Yes, Lucy! We did it! I'm flying!

Jimena: I'm flying! Just keep it steady, just ...!

...

The "bird" dives sharply into the water.

Lucy: Nine seconds.

Lucy: An improvement over last time.

Lucy: Do you need any assistance?

Jimena: Ugh, no, thanks.

She moves to unfasten the wings before noticing a crab perched on her shoulder. Returning it to the sand, she crumples down beside it and sighs.

Lucy: It appears the primary difficulty is power. The wind is not strong enough, or more precisely, we are not generating sufficient lift.

Lucy: With further optimizations, we should be able to extend your flight time.

Jimena: This model took a whole month to complete, and it only flew three more seconds!

Jimena: Is it ever going to work? Dad had to be crazy to spend his life like this.

A gaggle of boys nearby had watched her from their boat. They laughed as she fell, and are still laughing.

The girl shoots them an icy glare, a piercing bolt that hits them one by one until each wises up and turns back to their fishing.

Lucy: Three seconds represents a 50% improvement in performance. If you take that into consideration, our progress has been remarkable.

Jimena: Didn't take you for an optimist.

Lucy: Please, you should get yourself into some dry clothes. I believe I have identified the primary issues with the current design.

Jimena: Great. At this rate, we might make it from the cape to the shore in, oh, about a hundred years or so.

Jimena: *sighs* If only I could reach the other side of the ocean this very moment!

Jimena: Just think how it would be to fly all across the Atlantic and land on the shores of New England!

Lucy: We should head back.

Jimena: Coming.

Bundling up the soaked machine, the girl and the mechanical woman head back toward the city.

In the distance, the line of warships have resumed patrolling the harbor, their sails jutting up from the water like the fins of hungry sharks.

Port, Lisbon, 1835

As the midday hours turn toward the afternoon, the city begins to bustle with activity once more.

Jimena: So, what was the problem you talked about? The wind can't lift me? What does that mean? Am I too heavy? Do you think I'm fat?

Jimena: Have you ever seen kids playing with kites? All you need is a good running start, and the wind just takes them up far into the sky.

Jimena: How about we skip birds and just make a great, big, man-sized kite?

They walk together, talking and laughing; their doomed flight soon forgotten—the warmth and energy of Lisbon's streets lifting their spirits where wings had failed.

The mechanical woman, however, gives no response. She's replaying the experiment within her metal brain, analyzing it detail by detail.

Lucy: We need a pair of secondary wings installed 4 inches from the second joint.

Lucy: They would unfold after the primary set, in order to balance the lifting force of the wind across your body.

Jimena: Are you even listening to me?

The would-be pilot jumps into the mechanical woman's way, arms planted on her waist.

Lucy: The structure of a kite is extremely lightweight.

Jimena: So, you are calling me fat?

Lucy: I am merely asserting that a grown woman is much heavier than a toy made from wood and paper.

Lucy: A kite has a very limited payload, and it cannot be directed while in flight. Our model must be able to do much more than that.

An argument breaks up their conversation. From the slope ahead comes a stiff warning and a slurred response.

A squad of soldiers lead their latest "catches" out from the tavern, driving them away down the street.

Jimena: Looks like the rumor is true. They are hunting the smugglers' accomplices in the city.

Jimena: Or, whoever they've decided is an accomplice.

Lucy: It seems to have little bearing on our situation.

Jimena: Doesn't it? Just look at who they've arrested. Fernandes the Broken Arm, Carlos the Golden Hand, and Alfonso the Gill ...

The girl's face grows pale as she keens their faces out from the dark. All of them local arcanists, no exceptions.

Jimena: Of course, why should it bother you? You're not a local. Her Majesty would never dream of being so unkind to foreigners.

Jimena: Besides, you have that letter of introduction from that Computing Center. Who would be so stupid to offend an honorable guest from Britain?!

Jimena: But I'm not so lucky. If only I could fly, then if they ever came for me, I would laugh at them from the safety of the skies.

Lucy: There was no luck involved. The LSCC sent me here with all the necessary documents to accomplish their interests.

Lucy: I suppose I had not considered that arcanists here might not have the same privileges I do because of them.

Lucy: Returning to our work, where are the spare materials your father left?

Jimena: He kept them under St. George's Cathedral. It's on the east side of the city.

Carrying the still-dripping gear, the girl seems to forget her prior concerns, running ahead to lead the way.

Jimena: Keep an eye out for more of those soldados. We'll make a detour if we have to.

Jimena: Don't worry. We're going where the Queen and her dogs aren't likely to follow.

However, it seems Her Majesty's dogs have already caught their scent.

Lucy: Quiet.

Lucy: I believe that soldier may be an issue.

Jimena heeds the warning, passing the wet contraption over to Lucy.

Jimena: Shh. I'll meet you at the second junction past the checkpoint. Good luck!

She whispers her instructions and disappears into a narrow alleyway.

The sight of a mechanical woman holding a strange dripping machine cannot help but attract attention.

Soldier: Er, what have you got there?

Lucy: But if you must know, metal, wood, leather, fabric ...

Not to mention a badge bearing two lions that signifies an empire on which, it is said, the sun never sets.

Lucy: It is a device. Wings.

Wings.

The soldier's eyes bulge as he contends with the information, before signaling for her to move along.

The mechanical woman continues on unhurriedly toward her rendezvous.

Jimena: He didn't question you?

Jimena: What did you say to him? How come he just let you go?

Lucy: He only inquired as to what I was carrying, and I replied honestly: wings.

Jimena: ...

Jimena: Haha, I guess he took you for a weirdo and let you go. You aren't exactly what these dogs are hunting for anyways.

Jimena: Anyway, we'll reach the cathedral soon. It's in a spot underneath the pulpit. Well, if no one has cleared it out yet.

Lucy: Why would anyone do that?

Jimena: The poor are short of everything here. Metal for cookware, fabric and leather for clothes, and wood for their hearths ...

Jimena: They'll use whatever they can get their hands on. I just hope it's still there. I want to fly, to finish what he started, so he can rest, and so I can escape this place.

Lucy: We still have many problems to solve before we get there.

Lucy: A compiling set.

But these problems can be handled, shaped, like pruning the buds on a tree. Solving them one by one until we reach our goal.

Jimena: And it's taken us a whole month to solve just one. Maybe you've got that kind of time to spare, amiga—Not me.

Lucy: Your father spent ten years just to get us here.

Lucy: And he gave everything he had to this project.

Jimena: Yes. He did.

Jimena: Was it worth it? Will it ever be worth it? I know it was his dream and mine, but it feels impossible, or so far away, I'll never live to see it.

Lucy: When they first introduced the locomotive, people laughed at it. It was slower than a horse, and far less practical.

Lucy: Now, they talk of building railways to connect cities across the continent. Look at their progress. Imagine what might come of them in the future.

Jimena: And this hunk of junk that's flown a whole nine seconds—you think this is our future?

Lucy: I do.

She answers without hesitation. The woman herself is a product of this unstoppable revolution.

Jimena: Alright. Well, we're here. Let's see if we can find our way to this future with whatever is left of the past.

The two of them arrive at the ruins of a cathedral damaged in the war. A solitary statue of an angel remains, though it has lost its wings, and with them, it seems, all holiness fell away too.

Lucy: I do hope the materials are still intact.

Jimena: There is a cellar underneath the pulpit. Go on ahead. I need to speak to a friend around here. Be right back.

The Blown Away Kite

CH19601-02

The Blown Away Kite

Cathedral, 1835

It's quieter than she imagined.

Lucy: A kite.

Lucy: She may be right to consider that ...

After checking the materials are still in reasonable condition, the mechanical woman begins to reimagine their design with the elements of a kite.

Lucy: Widen the bottom.

Lucy: Narrow the top, and yes, we get a triangle.

A wider rigid base should improve their ability to catch the wind.

She writes down her idea and looks over the design draft that must have been left by Jimena.

Lucy: That would allow the air to pass above and below the wings to improve lift.

Lucy: But what if the wind's heading changes abruptly?

She writes down the question, pondering over the words a while, before drawing a long curve along the draft.

Lucy: A feather tail to maintain balance. That may work.

The mechanical woman begins planning out the construction, when an anxious voice breaks the silence of the room.

Jimena: Lucy!

Lucy: Jimena?

Lucy: Calm yourself. What is happening?

Jimena: They've locked down the whole port now!

An escalation, but not an unexpected one.

Lucy: I see. So? What does it have to do with us?

Lucy: They can stop ships, Jimena, but they cannot stop the wind.

Lucy: Let us just continue our work for now. Tomorrow, we will go to the beach and have another try.

The mechanical woman's determination is unchanged, even as her companion wavers. True flight must be achieved.

They are heirs to a legacy that began in sketches from that enigmatic Italian genius, passed down through centuries, now standing on the cusp of realization.

Jimena: Now there are even more soldiers on the streets. Lucy, I-I'm afraid ...

The future of their project hangs on a thread.

Lucy: What is it that you are afraid of?

Jimena: That they'll arrest me, Lucy! And only God knows what else, if I stay too long. That I'll never fly—Never reach New England.

Jimena: Ah, if I had only gone with Fernando when he left ...

Lucy: Are we just going to leave these materials behind?

Rough-wrought metal pipes, wood, coal, leather ... All kinds of materials prepared for the project, all lying hidden beneath this ruined place.

This cathedral is a scar, a victim of the civil war that had swept through Lisbon not a few years ago. And even in this relatively intact cellar, it is far from an ideal workshop.

Lucy: I believe that would not be a wise choice.

Lucy: Mr. Murdock is very supportive of your father's project, as is the LSCC.

Lucy: It was your father's letter that brought me here.

Lucy: We have had few ideas as groundbreaking as this in LSCC since Mr. Watt passed.

Lucy: James Watt. Some of my colleagues see him as my "father." I had been part of his steam engine before my awakening.

She remembers the strange thrill of her first moment of awareness, welcomed by a chorus of steam and whistles.

Jimena: Then I suppose he was in a way, don't you think? We're all built by our fathers, in one way or another. I remember how my dad lifted me up high into the air. He said that someday we'd fly into the sky together.

Jimena: It's a beautiful dream. But, he's gone now. He traded his life away for nine seconds.

Jimena: I'm not sure that trade will ever be worth it.

Lucy: Jimena, every second we have spent off the ground is a startling achievement. Not just for you and your father, but for all of history.

The little inventor frowns, weighing the costs of her dreams and her anxiety for the future.

Jimena: So then, do you think I'll be able to use this to fly away and escape across the ocean?

Lucy: No.

Lucy: For the foreseeable future, the only way across the Atlantic is by ship.

Jimena: And with the port locked down, there's really no way at all.

Jimena: Wait, there may still be a chance!

Jimena: I have to speak to someone; I'll see you on the beach tomorrow.

She scarcely finishes her thoughts before she's bounded up and out into the streets.

Lucy: Jimena!

That's what she has been like—a gust of wind—always moving, flying too fast to see what lies ahead.

Silence returns, broken intermittently by the rhythmic scratching of pencil on paper.

Lucy: A lighter alloy could lower the weight without reducing structural strength.

Lucy: Aluminum.

It would work perfectly, if only the material were not beyond unaffordable.

Lucy: No. Not worth further consideration.

Lost in her thoughts, Lucy manages to miss entirely the squad of soldiers as they enter the ruins and find her alone.

As she marks down measurements and theories, they at last announce their presence.

Soldier: You, woman, where is she?

Lucy: Could it wait for a moment? I am in the middle of some calibrations.

The soldiers surround her, yet she takes little notice.

Soldier: Please, senhora. We don't want to cause any violence under God's roof, even in its current state.

Soldier: We only need to know where the smuggler's daughter has gone.

Lucy: The secondary joints will be under the most stress, that will limit our maximum wing span. Unless, we might add an additional set.

She marks a spot on the draft with a circle.

Soldier: You're testing our patience, senhora.

A dagger pierces through the circle on the draft, pinning it to the table.

She raises her head to meet the man. Jimena was right. He bears a striking resemblance to a snarling dog.

Lucy: There is no one else here.

Lucy: Who were you looking for again?

Soldier: The smuggler's daughter!

Lucy: I am afraid you are what is the phrase: "barking up the wrong tree?"

Soldier: No more funny business, senhora. Give me her location, now!

Lucy: A logical impossibility. I cannot give you what I do not possess.

The soldier draws back the dagger roughly, ripping the draft apart.

Lucy picks the pieces up and calmly begins puzzling them together.

Lucy: Please leave.

Lucy: You are wasting your time.

She can sense their hesitation; their snarls starting to sound a little more like whines.

Lucy: All I have here is pencil and paper. There is nothing else, senhor.

Soldier: What about this, huh? You call that nothing else?

The young soldier seizes the winged contraption, then throws it to the ground.

Like a bull unleashed from its pen, he stomps over any breakable piece he can find.

Soldier: She's not here. Let's go! Vamos, rápido!

Her uninvited guests file out, tails between their legs, leaving the mess they've caused behind.

Wordlessly, the wings are picked up from the ground and placed onto the table.
It will be repaired, just as it had been many times before.

The mechanical woman draws out some spare materials and begins her work.

Icarus

CH19601-03

Icarus

Jimena: Phew! They're gone.

Lucy: Your father's materials proved useful.

Lucy: It will not take long to finish these repairs and make some adjustments. We will still be able to perform another test flight tomorrow.

Jimena: So, it's not all bad news. Still, I have something to tell you.

She stops, hesitating to continue.

Jimena: I'm leaving Lisbon.

Lucy: I see. And your flying machine?

Jimena: Leave it be. I'll take it and our drafts to the New World and continue the project there.

Lucy: So, you have no further need of me, then?

Jimena: That's not what I meant.

Lucy: If that means the project can be completed without further distraction, then I suppose I should welcome it. Congratulations, Jimena.

The mechanical woman stops her work abruptly and turns to her would-be test pilot.

Jimena: I've made a deal to secure my way out of here. I'm going to the beach; there'll be a boat waiting for me there.

Lucy: And you will be going to the New World? That would be unwise.

Lucy: You have access to many more resources here in Europe. To start over in New England would cause unnecessary delays.

Jimena: And you'd have me stay in Lisbon? Where I could be arrested at any time?

Lucy: So, you will leave? Even if that puts the project in jeopardy?

The mechanical woman's words sound like an accusation.

Jimena: I'll take all the materials with me, and our drafts. A delay, maybe, maybe a long one, but nothing I can't handle. I'll be fine.

Her confidence fades.

Jimena: Besides, this is my father's legacy, MY FATHER'S, not yours. You're only here because of him. This is my decision to make, it's my dream—our dream ...

The words taste like a lie.

She's scared.

Lucy: Your father built you a dream. Just like Mr. Watt's done for me.

Lucy: He wanted me to leave the laboratory to meet more people, to listen to them, and understand their ideas, their imaginations.

Lucy: He believed in the power of human cooperation. That an idea could be passed from hand to hand, like a torch or an ember, across generations.

She points at the core of her body, that very idea had come to life in her.

Lucy: Thanks in part to our cooperation, our project has already come so far from where it started.

Jimena: That's true. But things have changed. If there were some glimmers of success, something I could hold on to, maybe, but ...

Jimena: What's the difference between six seconds and nine?

Lucy packs the wings wordlessly and walks to the door.

Jimena: Lucy ...

Lucy: The wind is nearing optimal conditions. We should seize this final opportunity to test our work before you go.

Lucy: With our current technology, wind alone will not provide sufficient force to lift a human in a sustained or directed flight.

Lucy: But a controlled glide may be possible, with the right materials.

Despite knowing that they will soon part, the only thing they can speak about is their project.

Jimena: Like a windmill? Or a water wheel? Will it use steam, like your power source?

Lucy: Perhaps. A steam engine would generate a great deal of force, but it would be far too heavy to be feasible.

Lucy: There is one more option: arcanum.

Jimena: You mean arcane skill? No, absolutely not!

She goes pale, as if the idea had conjured some long-forgotten nightmare.

Lucy: I know. Arcanum could not be a long-term solution.

Lucy: It would be impractical to have an arcanist stand by for every flight, after all.

Jimena: I don't think you really understand what other arcanists go through in this world.

Lucy: What do you mean?

Jimena: As you no doubt heard from those soldiers, my family were smugglers. Or, really, all we did was predict the direction of the wind and sea conditions, for fishermen, sailors, and for smugglers, too.

Jimena: My father was guiding a ship out. He went with them this time, I guess, to buy some fancy new materials. But something went wrong. There was a storm. The survivors called him a "Jonah." They decided it was his fault.

Her face falls, looking stuck somewhere between anger and grief.

Lucy: Your father did not control the weather; he only predicted it. He should not have been blamed. That was ignorance, superstition.

Lucy: Like sailors that refuse to look back as they leave port. Their motions have no true impact on the safety of the ship. It is only a mistaken belief.

Jimena: You don't understand, Lucy. They do it because looking back means that you are not ready to leave. And the sea is quick to punish those that cannot face it.

Lucy: The sea does not punish. It has neither feelings nor intelligence. It cannot know, nor would it care, what the sailors do.

Lucy: Superstition is only a false comfort.

Jimena: But a comfort they'll still value against the vast and unpredictable sea. We can't help ourselves, humans and arcanists alike.

Lucy: But it has no value at all. How much effort and time is wasted on these useless ceremonies? It could be more efficiently used for securing ropes or advancing marine technology!

Jimena: A sailor can't do his job properly if he's thinking of home. If nothing else, people gain courage from their beliefs.

Lucy: They would be able to feel more courageous if they focused on gaining experience, and properly utilizing technology. Everything else is simply mechanistic fate. Luck, not belief.

Random luck at that. It couldn't be controlled, only computed.

But as luck would have it, at that moment, lights spring up from the other side of the beach, carried by soldiers moving swiftly toward them.

Soldier: There, on the beach!

Jimena: No! No, the boat is late.

She realizes as she says it: it isn't late; it was already gone, along with any other boat that has a chance of getting through the tightening blockade.

Jimena: Where are they?

The panicked arcanist scans her eyes across the sea, picking out distant shadows of sails speckled against the horizon.

They must be a mile out from her and gaining distance fast.

And behind her, marching boots thud over the sand, gaining ground just as quickly.

Lucy: There is still something we can do.

Lucy: Let me see the machine.

True scientific achievement will have to wait, but this will provide a temporary solution.

The light of arcane skill flashes as Lucy's piston starts to churn. She pours out a blast of steam, causing the wings to stiffen and change into an iron-gray metal.

Jimena: Wh-what is this? You never told me you could do anything like this!

Jimena: What did you do to it?

Lucy: This may be strong enough to keep you in the air, but I cannot make any guarantees.

Jimena dons the now metallic wings, flapping them like a fledgling eager to take off.

Lucy: Control your altitude, and aim as best as you can for the nearest boat.

Lucy: You must lie flat and keep your body parallel to the water.

Jimena: What if I fall?

Lucy: Then they may catch you, but you will be caught if you stay here.

Lucy is right. There is only one way out, and she must take it. Nervous feet find their pace. She runs toward the sea.

Extending her wings, she leaps with all her strength up and over the water.

Lucy: Arms open, body flat, and stay balanced.

Jimena: Lucy!

The wind lifts her up surely and steadily.

Jimena's stomach churns as she makes for the horizon, chasing toward the bare outlines of masts against the setting sun.

From the beach, there is no more sign of the little pilot. No way to know whether she landed safely. Against all odds, the mechanical woman finds herself hoping her friend did not look back.

A seabird skims over her from the darkness of the sea toward the lights approaching from the beach.

Jimena's flight is a little unsteady at first, but she flies. No one can take that from her now.

Lucy: She has flown the coop, gentlemen.

Soldier: And you just let her go?

Lucy: I am under no obligation to catch her. Besides, how could I?

Soldier: Ay! Find her! Swim if you've got to! Vamos!

The soldiers pay the woman no mind as they rush up the ridge to get a better view, nor does she mind them. Her little bird is long past where they could reach her.

Though arcane skill was not how either of them had envisioned achieving it.

The mechanical woman heads back toward the city walls, lost in thought, scarcely noticing the blaring horns ringing out across the port.

Lucy: We have tried so many designs.

Lucy: A variety of materials, shapes, structures, countless formula adjustments, and yet only arcane skill proved sufficient—a shortcut—one wholly unreliable in any practical application.

Lucy: They had made so much progress, only to part without a solution.

Jimena. That frightened little bird—she could not blame her, though she wanted to. All this so she could escape and leave her with nothing, not even a working theory.

Crackling shots ring out from the beach, the soldiers firing vainly into the sky as their captain rages. But Jimena is well beyond sight now, and the reach of their shot.

As if to bring her back to her senses, a tiny crab crawls out of the sand and snaps at her feet.

Lucy: Hmm? You want me to step aside, am I right?

Lucy: Your chitin is certainly harder than most things, but no match for metal. An arthropod of your size simply does not have the size or strength to damage me.

She catches the crab gently, scanning it for a moment before allowing it to scurry away into the sea.

It disappears beneath the waves, just as her friend did in the evening sky.

Never to come back.

Lucy: I let you cheat. But to find a real workable solution, there will be a price to pay.

Lucy: It will take time, determination, and courage.

Lucy: And now you chose to give up, Jimena.

Metropolis

CH19601-04

Metropolis

Victor's Workshop, 1908

Lucy: Victor, how much gasoline is left?

Victor: Enough for a few tests, but we'll run out by next weekend.

Victor: Don't worry. Our investors will send us another barrel before that. Still, we'd best use it sparingly. This stuff is expensive.

The engineer turns the page on his calendar, staining it with oil.

Several drafts are pinned to the wall, each marked with notes and proposed changes.

Lucy: This pipe will not be enough. It may warp if the temperature is not controlled.

Lucy: There is no guarantee of stable heat transfer. Perhaps we should add an outer casing to improve its strength.

Lucy: What do you think, Victor?

Victor: Oh, now you're interested in my input? Sure, you don't want to ask someone else—maybe one of your nut-job colleagues?

Lucy: The LSCC does not have any staff to spare for this project. I am afraid you will have to do it.

It has been six months since she heard about the engineer's project—the development of a true powered wingsuit—and dropped everything to join it.

Lucy: This fuel tank will be too small for a flight of any significant duration.

Lucy: Not that it matters. Our engine is already severely limited by our materials and weight requirements.

Lucy: But we can still give it a try. Our current improvements should meaningfully increase our flight time.

The mechanical woman turns to the wall and unpins several unneeded drafts.

Victor: Meaningful? I doubt we'll manage more than a few extra seconds.

Victor: They're expecting results. Real results. They aren't going to pay for a few seconds.

Victor: If the company pulls out, we'll have to terminate the project.

Victor: And they've been delaying pay day a little more each month. Think they're trying to tell us something?

Lucy: We have made progress. Our situation is not as grim as you make it out to be.

Lucy: Are you going to give up?

Victor: Give up? No way!

Victor: Do you know how much I've spent on this project already? Now, if I were smart, I would've never taken up this crazy scheme in the first place.

Lucy: There have been attempts before, Victor.

Victor: Yeah, I know. I remember the stories from when I was a kid. Granny flew like a bird over from Europe. Well, maybe she did figure it out, but she didn't think to write it down before she croaked.

Victor: And there have been plenty of others that have tried since! Did even one of them succeed? No! Not a single goddamn one!

Victor: So, here I am, about to be the latest one to make the same damn fool mistakes. Maybe the last.

A moment of bitter clarity strikes the engineer.

Lucy: The last one? Of course not.

Lucy: Why would you think that? The dream of flight—no, I cannot imagine humanity ever giving up on it.

Lucy: Are you aware this pointless debate has cumulatively wasted 3 hours 48 minutes and 19 seconds of our time—just this month?

Lucy: Time you might have spent better on checking these wings.

Victor: Don't tell me what to do, Iron Head.

The engineer walks away, muttering unclear but targeted complaints, until he pivots and returns back to the wings.

Lucy: Returning to my original query. I am concerned that heat from the exhaust will compromise the integrity of this piping.

Victor: Why not cool it down with water or something?

Lucy: A rapid shift in temperature would cause the pipe to explode.

Lucy: I admit that it is unlikely that this would occur during the course of our current maximum flight time.

Lucy: However, I suggest a change of material to avoid potential issues as we extend its duration.

Victor: That's not an option. We've only got what we've got.

Lucy: Why? We could greatly increase our efficiency, not to mention safety.

Victor: Because our investors asked us to! They want a model for mass production.

Victor: Our top priority is to give them confidence, otherwise they will cut our funding.

"Cut our funding"—a threat that's looming ever nearer.

Victor: Do you know how long those Wright boys are keeping their contraption in the air now? We're losing our only opportunity to make something of this project.

Lucy: The only opportunity?

Lucy: That is not true, Victor.

Lucy: Even if they were three to five years ahead of us, it would make no difference. We are heirs to the work of centuries.

Lucy: Besides, we are not precisely direct competitors.

Lucy: They have produced a rigid-wing aircraft, not a single-person wingsuit. There is room for both concepts.

Victor: Centuries?! Three to five years ahead of us?! How many three to five years are there before I'm dead and gone?

Victor: You're wrong, Luce. If they bring that thing of theirs to the market first, they win. In this business, winner takes all, pal! And loser gets nothing, literally nothing!

Lucy: Even if we are not commercially successful, our work will not be meaningless, as long as we advance scientific knowledge.

Victor: Yeah, maybe that's enough for your egghead pals at LSCC! But there's no glory in second place, Luce. No fame. No money. Not a red cent!

He seizes up, as if his rage is bubbling up under his skin, ready to explode.

Lucy: Calm yourself, Victor.

Lucy: Go get the car. We will have a test flight at the parking lot in two hours.

Victor: Didn't I already tell you not to bark orders at me?

Victor: Sorry, Ms. Lucy. I'm just frustrated. I'm on a short fuse these days.

Lucy: We should focus on our work, Victor.

...

The telephone rings.

Victor runs toward the phone, slipping over the slick oil. He's half-cursing as he picks up the receiver.

Victor: Dammit—yes, hi, hello, yeah, this is Victor speaking. How can I—

Investor: Mr. Victor, we deeply regret that we must speak to you, due to some recent developments with our competitors.

Investor: We've been forced to reassess your project, and we've come to the conclusion that it is no-longer economical for us to maintain our involvement.

Victor: "Economical"? What do you mean?

Investor: Simply put, we will be no longer providing any further funding or material support.

Investor: Given the growing success of your competitors and your own lack of results, it appears there is no longer a viable future for this project.

Investor: We appreciate your understanding in all of this. You will be hearing from us if we require your services in the future.

Victor: Hey! Don't hang up! What the hell are you saying?!

Victor: Hey! Hey!

Victor: Bastards.

Lucy: What is happening?

Victor: Got good news, and bad. Which one do you want to hear first?

Lucy: You look pale. Tell me the good news first.

Victor: The good news is we don't have to listen to those ignoramuses we called investors anymore.

Victor: The bad news is they're out.

Lucy: I see. So, they have abandoned the project?

Victor: In a manner of speaking.

Lucy: But we have not abandoned it.

Victor: ...

The engineer sits down on the oily stool, head in palms, inadvertently smearing oil over his hair.

Victor: I've got some friends to call. They might be able to salvage this mess.

Lucy: In the meantime, perhaps we should take a break. Would you like some coffee?

Victor: What? Coffee, nah ... thanks, Luce. But I'm not taking a break, this is it. It's over.

Lucy: Oh?

Victor: It was a fool's dream, Luce. Leave the flying to the birds and those damned Wright Brothers.

Victor: I'm talking about selling off all this junk. We might cover a part of our losses still.

Lucy: Are you that easily defeated?

Lucy: We still have sufficient materials to improve on our current design.

Victor: Oh, sure, and what about the one after that? Experiment after experiment, how many more times do I gotta fail before I wise up?

Victor: And what's the use? Even if we could do it, we're already too late.

Lucy: Come on, go get the car. We can run the experiment on the beach.

Lucy: No, actually, we should go to the Charles River.

Victor: What are you up to?

Lucy: The wind coming off the river will be steadier.

He moves to get up, but sinks down. Unsure of his next move, looking like a ship tossing in a storm.

Victor: ...

Victor: One last run then? For old time's sakes?

Lucy: It will not be our last.

Victor: Hmm, yeah. Let's hope. Anyway, you're right. The river is best. We'll have more attention there. Provided we get off the ground.

Lucy: This is the right decision. It would be irresponsible to give up before attempting all other possibilities.

She would not give up again. Her words conjure up an old memory—a friend from years ago in the distant past,

of the last time she had shared this path with someone, and how superstition and fear had taken it away from them, just before they reached their dreams.

And their obstacles here are far less significant. The only thing that stands in her partner's way now is his own ego and greed.

Lucy: Just think back on how much progress we have made.

Victor: Alright. We'll stick to the plan. But if we're taking the river, you'll have to fly. I can't swim.

Lucy: Not a problem.

Victor: Great! I'll get the car and call the reporters.

Victor: Let's hope we get the headline tomorrow.

"Air Farce"

CH19601-05

"Air Farce"

Charles River Bank, 1908

It's a beautiful day, and many people have gathered along the river to enjoy it. The spirit of this moment in time feels intensely hopeful.

Humans have conquered darkness with electric light and distance with the chug of locomotives. What will come next? Will they conquer the sky and beyond? If nothing else, they feel assured the future is bright.

Victor: Ladies and gentlemen, make way please!

Victor: Are the spotlights ready? I'll pay, Sam! Trust me, for once!

Victor: Thank you, sir. This won't take long, I promise. 30 minutes tops, and I'll be out of your way!

Victor has become one-part engineer and two-parts showman, frantically lining up equipment and impatient reporters at the same time.

Victor: I promise you, gentlemen, soon you will possess the most memorable and valuable photos of the century. Forget the headlines of The Sun, this is history—

Victor: Ah, pardon me. I need a moment to talk with my lovely assistant.

The engineer wipes off his fake smile as quickly as he turns, running over to help Lucy unload the metal frame.

Lucy: The assembly time should be approximately five minutes.

Victor: Yeah, yeah, great—I just got a brilliant idea, Luce.

Lucy: I am listening.

Victor: We both know we can't afford another failure.

The engineer glances around and hushes his voice.

Victor: Our current model still has some problems to be solved, so why don't you just use that arcane skill of yours to "spruce her up," you know?

Lucy: You mean you want me to change its mechanical structure to make the experiment work?

Victor: Yes, yes! E-exactly what I mean!

Lucy: No. I refuse.

Victor: What? Why? What's ...

Lucy: First, you must remember there are limits to my arcane skill. I may run out of time, and it is possible it will not work.

Victor: The odds are already stacked against us in the first place! What could it hurt, even if it fails?

Lucy: It would be impossible for me to enchant every suit once our model is put into mass production.

Lucy: Without even mentioning that it would be impossible to enchant them each time they were needed.

Lucy: Arcane skills simply do not work that way.

Lucy: Our goal is a stable, reliable, and practical model. Who would buy anything less?

She focuses on the future, not the present, hoping to appeal to the engineer's true desire.

Lucy: Now, you have my reasons. Please assist me with the wings.

Victor: Do you have that head of yours screwed on wrong? Or do you really not get that our project is done if we fail again?!

Victor: Luce, it's real simple: enchant the engine, go up and have a little fun in the sky, then come back down to greet the reporters when they come to take photos of you!

Victor: Our investors will change their minds. Hell, they'll beg us to continue. Then we'll have all the money we could ever want to make this hunk of bolts really work!

Victor: And once we actually finish it, Luce, buddy, we'll have stacks as high as Rockefeller.

Lucy: OK. But why would we want a pile of paper?

Victor: ...

Victor: That's all I've ever wanted! Ever since I found these old drafts while cleaning up my old nan's house.

Lucy: Your motivations notwithstanding.

Lucy: This project has a promising future. That is all I know.

Victor throws the drafts to the ground in a huff. Lucy picks them up and returns them to the car.

Lucy: We still need these.

Victor: Like hell we do!

The engineer storms away, but as he enters sight of the crowd, he puts on his fake smile as cool as he can be.

It's smaller than he'd hoped.

Lucy: Victor, please ask the crowd to disperse.

Lucy: There may be unexpected dangers. I do not want anyone to be hurt.

Victor: You heard the lady, folks. My assistant is meticulously careful and considerate. Even though our invention is completely safe.

Victor: Still, as a practical humanist and a thorough scientist, she's always taking even the rarest and most extreme worst-case scenarios into consideration. But I do really have to emphasize "rarest."

Victor: Now, ladies and gentlemen, let's witness the latest creation of Victor's Workshop: the fantastic wingsuit! Press one button and control the wings with a simple lever. That's all it takes to fly right up into the sky!

Lucy: Hold on, Victor. We are not quite ready.

Lucy: There is some turbulence in the air. We should delay for a short while until it passes.

Victor: ...

Victor: Excuse us, friends. I'm afraid we need to adjust our equipment just one last time to make sure it all runs smoothly. Please, give us just two more minutes!

The Crowd: Hey, what circus are you from, pal?

Victor: Victor's Workshop of Boston, sir. We're not a circus or some arcane swindlers.

The Crowd: A pair of iron wings? Installed on a robot's back? Sure sounds like a clown act to me! Haha!

The Crowd: Go on, let's see it then! What can she do?

Victor: We can't keep waiting, Luce. The crowd's turning on us.

Lucy: I am on standby until the right wind arrives.

Lucy: You appear anxious. Why is that?

A ship floats past them on the river, its passengers gawking at the scene.

Victor: Because, we need to take off now. Or all those very nice folk out there are going to leave, and our last chance for fame and fortune goes with them.

Lucy: Is that so? What does that mean for the project?

Lucy: What good does that spotlight do for our experiment?

Lucy: Will the people gathering by the river help us optimize our wingspan, our power source?

Victor: You're being difficult. What's the point of this experiment if no one is here to see it?

Lucy: The point? The point is that we put our work into perfecting this machine. Is that not point enough?

Lucy: Even if something does go wrong, it will give us a chance to analyze and fix it. That is the point.

Lucy: Humans have dreamed of flying in the wind since before even the first kite flew into the air.

Lucy: And now, our achievements in materials science and the development of the internal combustion engine have laid the foundation for us to make this dream a reality.

Lucy: What do we lose even if we fail again, Victor? We have failed a hundred times. What is so terrible about one more?

The engineer has no answer for her. The moment turns as a swift westerly wind blows across the river bank.

Victor: That's your wind, Luce! Go on!

Victor: Move! Move! Good luck!

Lucy: It is in the hands of fate now.

The mechanical woman pulls the device over her silvery arms.

Now a bird with iron wings, she walks to the edge of the river bank and presses a button as a jet of air erupts beneath her.

No signs of arcanum, only the mechanical rumble of human technology.

Lucy: We have done all we can. I have a feeling that this time we will succeed.

Victor: You ready?

Victor: Ladies and gentlemen, please direct your eyes—

Victor: For you are about to bear witness to the greatest invention in human history!

Victor: Where Icarus failed, we succeed. Now, mankind will take to the skies just as easily as the birds.

Like her friend ran toward the sea those many years ago, Lucy spreads her wings and leaps from the river bank.

The Crowd: Hey! That was a jump! She jumped!

The Crowd: Come on, do any of you think that stupid contraption is going to fly?

Where many expected a splash, they hear instead a roar, as the engine pulls her up and over the surface of the water.

Lucy: There appears to be some unexpected instability.

Victor: Look, look! She's rising, folks! That's no mere glide! But 100% genuine powered flight!

Victor: Up, and onwards!

The engineer stares out at the silvery bird as it holds just above the river's surface.

Sweat drips off his forehead, and the yellow draft paper in his hand is now a crumpled ball. But he turns back to the crowd.

Lucy: Victor—

Victor: Let me tell you folks. This thing is safe, stable, and reliable. Why even a child may pilot it after a few short lessons!

Victor: Imagine it, faster than an airship, more convenient than a locomotive, and cheaper than an automobile!

Lucy: Victor!

Victor: Gentlemen, I ask you now for just a small investment. One small payment, to put yourself in front of the most important invention of our time!

Victor: This is a once-in-a-lifetime opportunity to own a piece of the future! Seize it now! Fame and fortune await!

Lucy: Victor! Something is wrong!

The engine begins to sputter and cough, no longer moving forward. The buffeting wind pulls her into a tumble.

...

The Crowd: Your future just took a nose dive, pal.

The Crowd: Oof, look how she hit the water! That could've killed somebody!

Victor: ...!

The engineer turns white, slowly wrenching his head toward the splash.

Victor: No!

Victor: No ...

The iron bird pulls herself up to the river bank and drops her wings.

Lucy: I believe that I have found our problem.

Lucy: Our engine is not providing sufficient downward thrust to keep us in the air, even with the aid of wind.

Lucy: There are also many extraneous parts. We could significantly streamline the design to improve our fuel efficiency.

Lucy: If we can improve the overall power output—

Victor: Yeah, well, good luck with that, Luce. I'm out.

Victor: No one is going to invest after today—Not a penny.

Victor: It's a money pit, and I'm tired of digging it. No chance I'm going to see a profit, not in ten years, not in a hundred!

The mechanical woman dries herself and walks to the engineer as he mopes on the ground.

They deserve a better ending.

Lucy: So, you give up? We got this close, and you just want to walk away?

Victor: Seems like it, don't it?

Victor ends his last attempt with a heaving sigh. Pulling his hat down over his face, he walks out into the crowd and doesn't return.

Lucy: This is only another failed result.

Lucy: We have found another avenue for improvement.

Lucy: This is nothing more than a minor setback. An unsuccessful experiment is not a critical failure.

The crowd is dispersing now, and Victor has disappeared with them. Flying is a beautiful dream, but dreams do nothing for a man with an empty wallet and a growling stomach.

Lucy: ...

Lucy: You only ever saw what you could reach in one grasp, Victor.

Lucy: But it takes more patience than that if you really want to touch the sky.

But he is gone; the random movement of fate has left her alone once more.

But that will not stop her.

Lucy: Like many before him, he was terrified by the difficulties ahead and changed his course, like those mountaineers who turned from the Matterhorn long before they reached its peak.

Lucy: But I know there will be others—those behind him—that will climb higher than he had ever imagined.

Inherited Wings

CH19601-06

Inherited Wings

Floridian Branch of Space Center, 1969

Space Center Employee: On behalf of the Space Center, we really wanted to thank you for your service these past years, Madam Lucy.

Space Center Employee: I brought you a few brochures for you to look over with some great vacation spots. Sunny beaches, high-class resorts—you'll have lots of things to do.

She accepts the offered papers gracefully, but there is no real joy in it for her. A holiday resort is no different to her from a quiet office with an adequate power supply.

Space Center Employee: Oh, and I included the schedule for the moon landing broadcast. Didn't want you to miss it.

Lucy: Would you ring the Bay Shore High School for me? I need to talk to Ms. Joanna.

Lucy: I am sure she will be in her office now.

Space Center Employee: O-okay.

...

...

???: Hello.

Lucy: Hello, Ms. Joanna. I am letting you know I will arrive later today, before ...

She looks up at the clock.

Lucy: Shortly before dusk. Yes, see you later.

Joanna: Okay. I'll wait for you here.

Lucy hangs up without another word. Never one for formalities.

Lucy: I will need a ride to Bay Shore.

Space Center Employee: Sure. I'll arrange for a car to pick you up outside the center.

Lucy: How long will it take to get there?

Space Center Employee: Less than an hour, if the traffic isn't too bad.

Space Center Employee: And that shouldn't be a worry. The military has arranged a cordon all around the center.

Lucy: Good. I am glad to see that the Space Center remains as efficient as ever.

Lucy: You have prepared for everything with more than adequate redundancies. I am looking forward to witnessing this historic moment.

Lucy: I will also need my luggage and equipment brought to the car before I leave. If that will not be too much trouble.

Bay Shore Middle School, 1969

Adolescents, even of the older variety, have always confused Lucy. The patterns of their behavior unpredictable and frequently irrational.

As she arrives, she is instantly reminded of that discomfort. The students begin flocking to her.

Student I: Woah, a robot! Hey Jack, Annie, come check it out! A real-live robot!

Lucy: Please allow me through.

Lucy: Please.

The mechanical woman is surrounded by students, forming an impenetrable wall of questions.

???: Give her some space.

???: Is that how any of you would treat a guest?

Lucy: Oh, Joanna. You look healthy. Have you improved your diet and exercise regime?

Joanna: Is that robot for: "good to see you?" Well, I'm glad you haven't changed too much from hanging around with all those brainiacs without me.

Lucy: A healthy body and a clear mind will improve performance in humans. And even I need regular maintenance.

Lucy: Without it, I might experience unnecessary component failures—Or, sorry, I mean to say, it is good to see you, too.

Joanna: Alright then, as for all of you, I'm sure you are eager to hit the beach. Grab your things and get going.

Joanna: I need a moment to speak to our guest. Off you go.

Joanna's Office, 1969

Joanna: These are their trophies. I know it's not as glamorous as working for the Space Center, but teaching has rewards of its own.

The trophies are not inconsiderable, and by the reaction of her students, she has most certainly achieved her own kind of success.

Lucy: I am glad to see that you have found something meaningful to do after you left.

Joanna: And I'm happy to see you finally taking a break.

Joanna: I was beginning to think you'd be working until your wiring melted.

Lucy: Hmm? As long as my power supply is stable, there should be no risk of damage to my wiring.

As if matching her statement, the lights begin to flicker in the building.

Joanna: Probably just the generator. Don't worry your pretty metal head over it.

Lucy: Oh no! You should—

Joanna: Relax, Lucy. We've all gotten used to it.

Joanna: That Space Center of yours is a bit of a power suck. But a few flickering bulbs aren't a bad price to pay for the moon.

Joanna: It shouldn't affect the broadcast. But we'll just have to hope the system can hold up until then.

Joanna: But, that's why we always have a plan B, right?

Lucy: Agreed. People underestimate their value. But in an emergency, redundancies always prove their worth.

Lucy: But best not to leave it to chance.

...

Joanna: Found it.

Joanna: Remember this?

They unfold a tattered old draft. A gift for Joanna to celebrate her retirement. One of the few human rituals she has come to value.

Lucy: I am happy that our modern advances in aviation have, at last, made this project viable.

Lucy: But, with my other work, I have not had any time to work on it. I am glad you have been able to make use of it.

Joanna: Ah, well I had the time you didn't, but you're the one that's really driven this project forward.

Lucy: Attending your experiments and record-keeping are far from significant contributions.

Lucy: And it has been a long time since I have done even that.

Lucy: What is your current progress?

Joanna: Not as far as your rocket, for certain. But we're making real progress.

Lucy: Invention takes time. I cannot fault you for lacking the same resources I have access to at the Space Center.

Lucy: It is remarkable just how much progress we have made even in the last decade.

Lucy: And it has been an international effort, even despite our differences. It is remarkable how much we have learned from the Soviet Space Program.

Joanna: The competition starts as soon as someone takes the lead.

Joanna: The Baikonur Cosmodrome sent the first satellite above the stratosphere, and we sent the second one.

Joanna: Then they launched a manned rocket into space. We had to catch up with them, so we made our own.

Joanna: Now they're aiming at the moon, Mars, and maybe the galaxy is next.

Joanna: But we're not going to be left watching their exhaust this time. Now we're going to take the lead.

A prideful smile stretches over her face.

Lucy: I appreciate your competitiveness.

Lucy: It makes us work faster, think smarter, run farther.

Lucy: Once we leave those first footsteps there, in the dust of another world,

Lucy: there is no telling how far humanity will run.

Lucy: There will be new footprints, new detectors, new astronauts, new bases, and then settlers. The Space Center has made plans for all of it.

Her friend has turned back to the trophy case.

Joanna: Jack and Peter are always competing with each other. They've pushed themselves so hard, and each time they fly a little higher and a little longer than the last time.

Joanna: I think near half of these are Jack's, and most of the rest are Peter's. I doubt they'd have gotten this far if the other wasn't there to push them another step.

Their awards tell the story as well as she could, small incremental gains, iteration over iteration. But each victory was a badge of honor.

Lucy: So, you have them working on the project too?

Joanna: I brought them on board as soon as they showed interest. Their work is largely based on your designs.

Joanna: That prototype you sent us served as an excellent model.

Lucy: Is that so? It had been shelved for a long time.

Lucy: Every once in a while, I would take it out again, whenever I found a new technology or material that could be incorporated into the design.

Lucy: It has been the work of decades, centuries almost. I am very glad to hear you have made progress on it.

Joanna: Only because you never gave up on it.

Lucy: A project should not be idly abandoned, not until all avenues of success have been exhausted.

Lucy: And that is a rare case.

Lucy: In fact, perhaps I am optimistic, but I think that such a case almost never happens.

Lucy: The only real challenge is in what projects deserve our time and resources. There is no way to juggle everything. Sometimes other work must take precedent.

Joanna: That's why we share these ideas. What was it you always said? "Like an ember passed from hand to hand?"

Joanna: C'mon, Lucy. Let me show you what we've done with that ember.

It was an uplifting invitation.

Joanna: After two years, we're finally seeing the sparks take flame. Though it isn't much now. Just something for them to play around with.

Lucy: You mean, you have actually made something of my designs?

Lucy: How did you overcome the issues with fuel capacity?

Lucy: Were you able to find a rigid material capable of withstanding the drag force? One that is still flexible enough to allow the pilot to pivot in the air?

How long had she kept that ember? How many before them had thrown it away? But now, it seems that it is burning once more.

Joanna: ...

Joanna: Ha-ha-ha-ha!

Joanna: Oh, Lucy, I'm sure you have a million questions. But let's just go see our pilots in action.

The Fledglings

CH19601-07

The Fledglings

Beach, Bay Shore Town, 1969

Birds swoop out over the water, flying east over the crashing waves.

Joanna: You know, the papers love writing crazy headlines about Florida.

Joanna: Like "Ghost Ship Returns from the Atlantic," "Lost War Plane Guided Home by Little Green Men," "Winged Man Lands on Tampa Beach" ...

Lucy: I have heard them before. Our colleagues sometimes share their favorites over lunch.

Lucy: They would laugh so hard they spilled their coffee.

Lucy: It was always a mess afterward.

Joanna: They say it's the "Devil's Triangle" three points between our coast, Puerto Rico, and Bermuda.

Lucy: I never considered them credible.

Lucy: The waters in this so-called triangle are no more treacherous or magical than any other. People should be careful about separating truth from superstition.

Lucy: I do not doubt that you are more than capable of that.

Joanna: Of course I can, but, I mean, that last story, I guess I want it to be true. What if it is?

Lucy: I see. So, you believe that a winged human appeared from the Atlantic and landed here?

Joanna: Well, it's just that after you gave me that draft from Uncle Victor.

Joanna: At first, I thought we'd never get any further than he did, but when I saw that, I thought maybe we can do it. Maybe it was fate?

Fate. Fate has no direction in her view. Ascribing purpose or intention to a chain of events was just a flaw in human reasoning.

Lucy: I never considered that.

Lucy: I had insufficient time to work on it.

Lucy: So, I gave it to you. That is all.

Lucy: Where your uncle failed, I had hoped you might succeed.

Lucy: There was no notion of destiny. I simply made a practical decision.

Lucy: And this winged man story is just sensationalist journalism. It was a coincidence.

Joanna: Say what you will. You may not believe in fate, but plenty of others, humans and arcanists alike, do.

Joanna: Anyway, fate or not, you should see how far we've come.

Student II: That makes it one minute and eight seconds! I win!

Student I: Give me a break. That was one minute and one second! You were still on the ground when you started counting!

Student II: Yeah, because we agreed to start counting as soon as the engine starts!

Lucy: That would be unwise. It is more useful to count only after you have achieved lift-off.

Lucy: Otherwise, your records may not accurately show your progress, sending you down the wrong path.

The boys turn to the source of the stern yet encouraging voice.

Student I: Wait, it's that robot from before! Cool!

Student I: Hey, everyone! Ms. Joanna brought the robot here!

Lucy: Do you mean that your flight duration now exceeds a minute without aid from arcane skills? Simply astounding.

Joanna: You should be saying thanks. It was Ms. Lucy who gave us the original prototype for our wings.

Student I: Of course, thanks, Miss!

Student II: Yeah, thanks, Miss Robot!

Joanna: You've correctly guessed our biggest obstacle: fuel.

Joanna: We tried enlarging the tank, but the heavier the weight in fuel and material, the less efficient it became. We just can't find a solution that allows for prolonged independent flight.

Joanna: But we're still confident, aren't we, gentlemen?

Joanna: It's far from perfect now, but that's fine. As you said, it's only a matter of time, and these kids have plenty of that.

Joanna: Think how far they'll have progressed by the time they reach my age. The world could be completely different, even from our wildest dreams.

The march of progress has been undeniable, even contained to just the last century. From whale oil lamps and coal, to gasoline and atomic energy, it is hard to imagine what will come next.

Lucy: The television is a great example. It has only been around for 44 years. When it first released, it seemed as though it would never dethrone the cinema.

Lucy: Now it is a staple of modern life. People can hardly spend a day without their favorite shows.

Student I: Are you talking about TV shows? Hey Miss, do you know Captain James T. Kirk and Spock?

Student II: Why did you ask her? It's a TV show, not real life, doofus. She's a robot, not an actress!

Student I: Yeah, I know she's a robot! That's why I figured she'd have to know about the best sci-fi show of all time.

Joanna: Let's stop pestering Miss Lucy about her favorite shows, alright boys? Unless, you'd like to spend tomorrow cleaning out the school swimming pool?

Joanna: Sound like fun to you?

It is a half-joking threat, but enough to bring the boys in line.

Lucy: That would be quite a shame. You would very likely miss the broadcast.

Lucy: I am very proud to say it is LSCC technology that was pivotal in constructing the signal-boosting devices.

Lucy: But for now, let me see the problem that you have been dealing with.

The mechanical woman walks toward the metal construct lying in the sand.

Student I: Daniel and Annie and the others helped us move it out here.

Student II: Yeah. We promised them everyone would get a try after we've ironed out the kinks.

Joanna: We've hit a hard limit now; I've tried everything I could think of to help them.

Lucy: One minute is already a remarkable achievement. Is this why you invited me here?

Joanna: Well, that and I miss working with you, Lucy.

Joanna: The switch was worth it, believe me. Still, I do miss some parts of my career, and my colleagues.

Joanna: So, think you can lend us a hand?

Lucy: I will attempt to do so. Is it still powered by liquid fuel?

Joanna: Until recently it was. We've actually been getting it straight from the Space Center. So, it was about as high-grade as you can get.

Joanna: But, we've been toying with something else now. To cut the weight down, we switched to wired electricity.

Lucy scans over the beach, finding a long black cord connected from the machine back toward a nearby generator.

Lucy: A wire? That completely restricts your flight range.

Lucy: The length of the cord would determine your maximum radius.

Lucy: ...

Lucy: You have created an electric kite.

Joanna: At least we replaced the thread with a wire.

Joanna: But you're right. It's nothing more than a kind of powered kite. They even call it that sometimes. But what else can we do?

Joanna: This was the only solution that gave us any kind of stable flight.

Joanna: Once unplugged, the battery can only sustain the engine for less than a minute.

Lucy: I see. So, it is not a matter of fuel, but rather of capacity.

Lucy: A simple adjustment will resolve this issue.

She produces a silvery metallic block from her coat, an emergency power source, made from the miraculous fusion of arcanum and material science.

Lucy: If your design has a T-2 general slot, this battery may work as a substitute.

Lucy: Let me try.

She skillfully replaces the battery.

Then unplugs the wings.

Lucy: Try it again.

Student I: So, what is it? A new battery?

Student I: Is that going to be enough?

Lucy: I am confident it will dramatically improve your flight time.

Lucy: However, I suggest you do not fly too high or too far away from the beach, at least until we know the precise power draw.

Joanna and the students crowd around the wings to see the new component.

Joanna: I didn't know the Space Center had been working on battery technology.

Lucy: Actually, this is a product I developed with the LSCC in 1965.

Lucy: An arcanist colleague there helped me complete this project. The arcanum imbued in it should allow it to ignore the usual physical limits of its material.

Joanna: I can only imagine what could be done with technology like that. Has it been released?

Lucy: We have begun mass production. We are only waiting for official permission to make it publicly available.

Lucy: By harnessing arcanum, we were able to achieve a similar energy density to rocket fuel, but without the need for cumbersome tanks and piping.

Lucy: Petroleum was a marvel of science that laid the foundation for modern civilization. That being the case, I believe that if arcane skills could be studied comprehensively and replicated, they might be capable of supplanting fossil fuel entirely.

Joanna: I always thought you refused to use arcane skills to enhance technology?

Lucy: I opposed the use of arcane skill only because it was unstable and not replicable at scale.

Lucy: However, if it can be incorporated into technology in a controllable and generalized way, it can work.

Lucy: The gap between arcane skills and human technologies is really not so different, no more than that between a human brain and a computer.

Lucy: One is created through genetic coding, while the other is created by digital coding. They are essentially the same.

Joanna: ...

Joanna: If it were anyone else saying that, I'd have called them a nut job.

Lucy: Why?

Lucy: A fact remains a fact, regardless of who says it.

Lucy: And a mechanical device has the same function, no matter its user.

Joanna: That much is true. But we're getting ahead of ourselves. We still need to see the results of our test flight.

In a short time, the first brave pilot takes off on his aluminum wings. The students cheer as the stopwatch counts up to 60 seconds and then beyond.

Lucy: It seems like a remarkable success.

Joanna: This will be the end of an era. No more competition. Just fun.

Lucy: I think we can call our project completed.

Joanna: Well, only thanks to you.

Lucy: Do you want me to report this achievement to the Space Center? Or maybe the LSCC.

Joanna: Not quite yet. There's still room for improvement. So how long do you think that battery can last now?

Lucy: We will need to do further testing to determine that; however, I suspect it is an improvement by orders of magnitude.

Joanna: Then maybe I should call him down. Just to be safe.

Joanna: Jack, get down here. Danny, Peter, please clear the beach off for landing.

Reluctantly, the young man returns from the sky, making a steady landing on the sand.

Lucy: Ten minutes and 48 seconds. That makes for quite a new record.

Joanna: I'm certain they'll have beat that handily before you know it.

Roaming through 1969

CH19601-08

Roaming through 1969

...

Lucy: Joanna.

Joanna: It's open. Come in.

The mechanical woman enters her friend's house.

Joanna's Home, 1969

Lucy: Did you move the TV upstairs?

She stands in a tidy living room, but there is no sign of her host.

Joanna: Yeah. The antenna's stuck. Michael couldn't fix it, so we brought it up here, where it could catch the signal.

Joanna: Come on up, Lucy. I'm trying to get this darned thing working. It won't take long, I promise.

Lucy: Are you in need of help?

Joanna: It's okay. I'm almost done. It's just this screw. It won't get in there—don't make me get rough with you, you naughty garoto!

The mechanical woman sits down and looks at the clock. It is 4:11 in the afternoon. She can easily imagine how busy her colleagues are at the Space Center.

Lucy: They should be close to landing on the moon now.

Joanna: Yeah, close. I'm close to making it work. Done!

The wires spark and burn. It strikes the woman like mocking laughter.

Lucy: It appears to be broken.

Lucy: A simple adjustment will resolve this issue.

Joanna: Alrighty then! Let's grab something to eat first, and then we'll see what I can do, or I guess, what you can do.

Joanna: Would you like to try some pamonha? It's an old family recipe from Portugal.

Lucy: I do not need food to replenish energy, Joanna. But, thank you.

Joanna: Oh, I completely forgot, haha.

Joanna: But it can't hurt to have something to eat, right? You are capable of it after all. Anyways, it's your call.

Lucy: Biological material has a much lower conversion efficiency than pure electricity.

It had been a century since she had changed her primary power source. It had increased her capacity exponentially.

Yet it came with an unexpected price: she no longer felt welcome at meal times. Humans consider fuel consumption a social activity, and some take offense to her abstaining.

Lucy: Perhaps high-nutrient foods will become more popular in the future. That way, less time will be wasted on social consumption rituals.

Joanna: Oh please, food isn't just about the health of your body. It's good for your spirit too! It's one of the great joys of living.

Lucy: Irrelevant. Whatever a spirit is, it is not nourished by biological matter. Now, consider the food we prepared for the astronauts, lightweight but densely-packed with nutrients—

Joanna: Fine, no pamonha for you. Now would you fix this machine? I don't want to miss the broadcast.

Lucy: Yes. It is possible. However, I must warn you, there is a small chance of failure. I predict between 2 to 5%.

Joanna: Well, I already called the repairman. But if you don't mind trying your luck, maybe you can give it a whirl first?

The mechanical woman reaches for the stubborn screw, and steam pours out from her core. The light of arcane skill seems to awaken the machine, and then ...

In a fit of sputtering smoke, the unlikeliest probability seems to have come to pass.

Joanna: *coughs* What's happening, Lucy?

Lucy: My arcane skill has failed.

A finger-sized robot now stands motionless next to a piston where Lucy had been. The piston squeaks up to Joanna from the ground.

Lucy: It has been 42 months and 18 days since the last time this failure occurred.

Lucy: ...

Lucy: How unfortunate.

Joanna: Seems like we made a bad gamble, haha.

Joanna: But it isn't so bad. Why, you almost look like a little dolly now. Very, very cute, Miss Lucy.

Joanna: But, how do you get back to your usual size?

Lucy: It will be over in just a few minutes. Do not be concerned.

Joanna: I never expected to learn something new about you after all these years.

Lucy: I am no more than a metal piston truly. Everything else is just a mechanical construction. A facade.

She looks now just as she did when she was first awakened.

When she took form as a part of that very first steam engine.

A few seconds pass, and Joanna picks up the tiny, immobile robot.

Joanna: I still can't believe this little thing was you.

Lucy: An accident caused by a rare instability with my arcane skill.

Lucy: I suppose you see now why I am reluctant to use my skills, even when they are important. It is dangerous to bet on an uncertainty.

Joanna: Thankfully it's only a TV this time. Not that big a deal, right? Besides, it was only a tiny, little accident. You can always try again.

Once again, Lucy emits steam from her core, this time successfully reactivating the television with her arcane skill.

Lucy: Done.

A signal reaches the TV from 300,000 kilometers away. On the black-and-white screen, a man begins to step out from a capsule, his feet landing on the surface of the moon.

The camera captures not only the capsule and the astronaut, but the silence and loneliness of space.

Joanna: It looks so beautiful up there.

Joanna: In some many different cultures, people believed something was living on the moon. In China, they say there's a palace on the moon, with a princess and a rabbit.

Lucy: There is no breathable atmosphere on the moon. Such a thing would be logically impossible.

Joanna: Well, we're about to see. They're taking the first steps now.

It may only be one small step for man, but for mankind, it is a journey of many thousand years.

She watches as the astronauts take their first clumsy steps across the moonscape.

Lucy: It was less than 70 years, Joanna, when most people believed humanity would never fly. And now, you have taken the first step on another world.

Joanna is still silent in awe of the moment, but in a short while, she points over to a painting on the wall.

Joanna: That painting was from Uncle Victor.

A painting that has kept her company throughout her life.

Depicted on it, a winged girl taking off into a harbor, flying past a lighthouse, heading toward the sun.

Joanna: They say she's an ancestor of our family, from Portugal. That she was the first human to truly fly.

There was no question in Lucy's mind of who this mysterious woman had been. It was a remarkable likeness of the scene.

Lucy: I started this project in 1835 in Lisbon. Jimena and I made the first prototype of those wings.

Lucy: There was no engine in the first designs. We could only catch the wind.

Lucy: It was something of a bitter moment—I had to use my arcane skill to alter it so that Jimena could reach her ship. I never saw her again. I did not even know if she survived.

Joanna: ...

Joanna: I knew she had a partner, somehow, I never realized it was you.

Lucy: I spent a lot of time on other work after that. By the time I heard about your uncle's project, it had been 70 years.

Lucy: But still, we could not solve the problem of power. The technology just was not available to lift the frame for any significant length of time. He gave up.

Joanna: Did you know after he left Boston, Uncle Victor went to New York, made it rich with some olive oil business?

Lucy: No, I never knew. But I am happy for him.

So, he achieved his goals after all. Just like Jimena did, in a way.

Lucy: He had always realized the business value of the wings.

Lucy: But he came to believe that it was impossible to achieve the design with the technology we had available.

Lucy: Humans have so little time, when you look at it. The future is always slipping out of their grasps.

Lucy: It is understandable why they grow frustrated after failing again and again. They do not have the time to wait around for the next miracle.

Joanna: But by handing down that ember, through you, he helped us get here. Now we've realized his dream, and Jimena's, and her father's before her.

Lucy: Yes, ideas, too, have a process of slow iteration.

Lucy: In the 19th century, only a few people even dreamed that humans would ever fly.

Lucy: In your uncle's time, it had already changed. People knew it was possible, but thought it would only ever be a novelty.

Lucy: Now, no one doubts where we can go.

Lucy: It is always only a matter of time, Joanna.

And time moves all, like a river. It cannot be stopped, and no one can control where it will take us.

No matter whether you rage against it, take every moment you can, or give up halfway.

Lucy: All that we need is patience.

Joanna: We will eventually get there, even though many of us may fall along the way.

Joanna: But you witnessed it all, Lucy. You kept that ember glowing.

Lucy: I just kept you company along the way.

Lucy: That is all I did.